Method	bigcode bench		champ		logicbench		medcalc bench		theoremqa		toolqa		Average	
	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10
BM25														
BM25	0.610	0.544	0.423	0.410	0.158	0.220	0.315	0.462	0.524	0.659	0.070	0.271	0.302	0.410
LLM Select	0.504	0.534	0.462	0.439	0.202	0.309	0.533	0.633	0.695	0.791	0.507	0.595	0.491	0.572
SkillRouter	0.358	0.292	0.423	0.410	0.228	0.357	**0.945**	**0.953**	**0.810**	**0.843**	**0.600**	**0.696**	0.603	0.645
SkillDAG	0.610	0.589	0.423	0.442	0.158	0.315	0.309	0.461	0.524	0.663	0.070	0.303	0.301	0.442
Skill-Pro	0.244	0.464	0.231	0.358	0.149	0.262	0.679	0.776	0.438	0.643	0.465	0.554	0.416	0.549
RouterBSI_init	0.431	0.447	0.308	0.367	0.158	0.281	0.642	0.720	0.448	0.627	0.256	0.413	0.384	0.495
RouterBSI	**0.878**	**0.729**	**0.500**	**0.498**	**0.281**	**0.371**	0.818	0.842	0.638	0.748	0.577	0.665	**0.640**	**0.675**
Vector														
Vector	0.504	0.407	0.346	0.463	0.105	0.248	0.988	0.992	0.695	0.833	0.312	0.413	0.516	0.575
LLM Select	0.496	0.432	0.308	0.459	0.237	0.337	0.861	0.945	0.686	0.839	0.363	0.463	0.519	0.598
SkillRouter	0.350	0.272	0.462	0.515	0.272	**0.382**	0.988	0.996	**0.819**	**0.866**	0.549	0.657	0.606	0.651
SkillDAG	0.504	0.544	0.346	0.469	0.105	0.364	0.988	0.990	0.695	0.830	0.312	0.563	0.516	0.658
Skill-Pro	0.220	0.360	0.231	0.432	0.096	0.248	0.715	0.892	0.419	0.726	0.335	0.451	0.372	0.540
RouterBSI_init	0.382	0.338	0.269	0.428	0.184	0.291	0.970	0.985	0.581	0.747	0.298	0.442	0.481	0.564
RouterBSI	**0.756**	**0.620**	**0.577**	**0.611**	**0.289**	0.367	**0.994**	**0.997**	0.667	0.819	**0.614**	**0.673**	**0.678**	**0.707**
SkillSight														
SkillSight	0.724	0.616	0.308	0.429	0.167	0.406	0.830	0.903	0.733	0.852	0.102	0.390	0.471	0.609
LLM Select	0.439	0.563	0.385	0.475	0.360	0.536	0.842	0.924	0.667	0.828	0.442	0.582	0.547	0.678
SkillRouter	0.390	0.332	0.462	0.503	0.307	0.509	**0.988**	**0.996**	**0.819**	**0.870**	0.651	0.782	0.647	0.716
SkillDAG	0.724	0.668	0.308	0.449	0.167	0.455	0.830	0.905	0.733	0.849	0.102	0.398	0.471	0.628
Skill-Pro	0.252	0.513	0.192	0.394	0.237	0.442	0.721	0.883	0.524	0.768	0.405	0.576	0.433	0.634
RouterBSI_init	0.585	0.465	0.462	0.510	0.211	0.449	0.879	0.927	0.648	0.804	0.316	0.505	0.520	0.625
RouterBSI	**0.862**	**0.771**	**0.577**	**0.558**	**0.439**	**0.608**	0.933	0.960	0.752	0.857	**0.702**	**0.797**	**0.742**	**0.800**

Table 1: Skill routing performance comparison using Qwen3.5 4B with BM25, Vector, and SkillSight. Columns show test nDCG@1 (N@1) and nDCG@10 (N@10) for each benchmark and the task-weighted average. Bold and underlined values indicate the best and second-best scores within each retriever group.